Authors:

Matteo Diano[a,b], Tommaso Ciorli[a], Ilaria Stura[c], Valerio Bioglio[d], Alessia Moiso[a], Francesca Persiani[c], Radu Covaci[a], Cristiana Angelini[e], Ennio Bilancini[e], Andrea Santoro[f], Lorenzo Zaffina[h], Giovanni Petri[g], Federico D'Agata[b,c].

Affiliations:

[a] Department of Psychology, University of Torino, Torino, 10124, Italy

[b] Neuroscience Institute of Turin, University of Torino, Torino, 10124, Italy

[c] Department of Neurosciences, University of Torino, Torino, 10126, Italy

[d] Department of Computer Science, University of Torino, Torino, 10149, Italy

[e] IMT School of Advanced Studies, Lucca, 55100, Italy

[f] ISI Foundation, Torino, 10123, Italy

[g] Network Science Institute, Northeastern University London, London, United Kingdom. Department of Physics, Northeastern University, Boston, Massachusetts, United States of America.

[h] Department of Computer Science, La Sapienza University, Rome, Italy

Title:

Higher-order models for fNIRS hyperscanning in complex and ecological interactions

Abstract:

Hyperscanning makes it possible to watch brains "move together" while people interact, capturing moments of alignment that would be not detectable if we studied individuals in isolation. Yet, much of fNIRS hyperscanning still treats social interactions as intrinsically dyadic, and summarizes coordination with bivariate metrics. In real social settings, though, coordination often belongs to the group: patterns can emerge only when three or more people co-regulate attention, timing, affective syntonization and shared meaning. To address this gap, we introduce a higher-order analysis pipeline for group hyperscanning and apply it to an ecological role-play paradigm in which participants pursue their own motives while simultaneously negotiating a shared storyline and team dynamics.

In our protocol, four players (40 players in total) and a game master engage in a structured RPG session designed to gradually increase interactional richness. The

session begins with relatively constrained synchrony contexts (motor pacing and joint movie viewing) and then moves toward open-ended role-play, where information is distributed through narrative hints and spontaneous dialogue. Neural coordination is tracked over time using sliding windows (20s, 1s step), benchmarked against circular-shift nulls and assessed with Bayesian Crawford t-tests after standard preprocessing (including TDDR prewhitening, short-channel regression, and band-pass filtering).

To capture complementary facets of “togetherness”, we combine intersubject correlation (ISC), homological scaffolds that quantify higher-order loops (H1), and O-information that disentangles redundancy from synergy. In passive tasks, ISC largely mirrors redundant coupling, consistent with shared stimulus-driven processing. During active role-play, however, the measures separate: redundancy follows broad shared synchronization, scaffolds reveal evolving higher-order structure, and synergy remains distinct, pointing to genuinely collective dynamics that cannot be reduced to any pairwise interaction.